Data-centre queues against flow-based import limits in the Norwegian bidding zones: a preregistered descriptive analysis

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Abstract

We ask in how many market time units the data-centre queue in a Norwegian bidding zone, added as load in that zone, would require more import than the published minimum net position allows. The headroom is the realised day-ahead net position minus the minimum net position from the Nordic flow-based capacity calculation, adapted from the import-headroom definition of Xu and Stiewe [10]: we use the signed difference NP − minNP and keep negative values, where Xu and Stiewe use the absolute difference. The analysis covers 2024-10-30 to 2026-10-31 for the five Norwegian zones and the share f of the queue on a grid of 101 points. A secondary measure adds the regulable hydropower that was not producing, and is an optimistic upper bound. Methods were preregistered in a signed git tag and submitted to OSF Registries before any outcome was computed [14]; three deviations are reported. At $f = 1.00$ in the primary scenario, the number of hours with an event over the whole period ranges from 53 (NO5) to 4984 (NO4) for the primary measure, out of 17567 valid hours. The results are descriptive; they make no claim about actual security of supply, prices or connection capacity.

1 Introduction

On 09.11.2026 Statnett announced: "Statnett innfører en midlertidig stans av ytterligere reservasjon av nettkapasitet til datasentre" (Statnett is introducing a temporary halt to further reservation of grid capacity for data centres) [1]. The halt concerns further reservation of grid capacity; according to Statnett, the decision does not mean a general halt in connecting data centres, and existing connections are not affected [1]. Statnett gives the systemic properties of data centres as the reason [1]. Statnett reports the data-centre queue per Norwegian bidding zone as reserved capacity and capacity queue [2]. The question we ask is the one in the preregistration [14]: the share of market time units (MTUs) in which the data-centre queue, added as load in the zone, would require more import than the minimum net position allows, per Norwegian bidding zone, as a function of the share f of the queue that is added.

Our earlier analysis, KHEPRI-DC v1 [13], added the queue as flat load to observed production minus load, but did not test import. Its interpretation rules treated import as normal operation without testing any limit [13].

Positioning. No page in our source archive as it stood after the first source reading (K-01–K-33; all sources are listed with URL, retrieval time and sha256 in the supplement `sources.csv` [15]) has data-centre terms and net-position terms together, apart from v1's own files. The closest are:

- Statnett's reply to NVE [3], which uses a fixed realised import capacity into the Northern area of 1050 MW and does not use flow-based terms;
- Xu and Stiewe [10], who define the import headroom but use data "only published for the Core FBMC region" and have no Nordic zones;
- KHEPRI-DC v1 [13], which does not test import.

We do not claim that the question is unanswered outside this archive.

2 Data

All series are placed on the MTU of the JAO data with duration weighting and without interpolation of missing values. The period is [2024-10-30T00:00Z, 2026-11-01T00:00Z), that is, including 2026-10-31. JAO business days are Norwegian days, which start at 23:00Z in winter time and at 22:00Z in summer time (the handbook gives the summer-time example [5]). All time stamps are interval starts in UTC.

2.1 Realised net position: JAO fbDomainShadowPrice

The realised day-ahead net position NP per zone is the field `flowFb` of the rows with `cnecName` `NetPosition_<zone>` in the JAO Nordic dataset `fbDomainShadowPrice` [5] (deviation A2, Section 4). According to the JAO handbook, the Net Position SDAC page publishes exactly this Flow_FB value of the NetPosition CNECs [5]. The endpoint is `https://publicationtool.jao.eu/nordic/api/data/fbDomainShadowPrice` [5]. The dataset has data from the first flow-based delivery day, 2024-10-30, and none for the business day before. All 733 business days were retrieved, each with a manifest entry, a check string and status OK. Of these, 159 days were taken over unchanged from a reconnaissance retrieval on 2026-11-10 between 04:50:35Z and 04:57:34Z, and the remaining days were retrieved between 05:14:34Z and 05:28:08Z the same day. The retrieval used jao-py 0.8.7 [7] with its own user agent and no pause between calls. JAO answered with HTTP 429 "Too Many Requests" four times, on three business days (twice for 2025-02-07); each of these days was retrieved on a later attempt, and all days ended with status OK. Whether such retrieval counts as use of bots under the JAO terms [6] is an open point. In the overlap with the JAO `netPos` dataset, the two are identical: 75 380 MTU pairs and 0 deviations. `netPos` itself has no data before 2026-05-28.

2.2 Minimum net position: JAO maxNetPos

The minimum net position minNP per zone and MTU is the field `min_<zone>` in the JAO Nordic dataset `maxNetPos`, `https://publicationtool.jao.eu/nordic/api/data/maxNetPos` [5]. All 733 business days were retrieved with status OK, on 2026-11-10 between 04:11:31Z and 04:27:39Z. Each minimum or maximum net position is "a unique extremum of the domain" [5].

2.3 The JAO MTU shift

By the preregistered rule, the MTU shift is the first business day on which every MTU is 15 minutes, applied to each JAO dataset, and the later date is the boundary. Both `fbDomainShadowPrice` and `maxNetPos` shift on 2025-10-01, so the boundary is 2025-09-30T22:00Z. Before the boundary, MTUs are 60 minutes; after it, 15 minutes.

2.4 ENTSO-E data

Installed regulable hydropower is taken from ENTSO-E A68 (Hydro Water Reservoir and Hydro Pumped Storage) [11], using the value for the calendar year in which the MTU lies (Table 1). Hydro Pumped Storage is "n/e" in NO1 and NO4 in all three years [11], and the pumped-storage term is set to 0 there. Actual production is taken from ENTSO-E A75 for the same two types [11]; A75 "Actual Consumption" for Hydro Pumped Storage is used only to decide whether an interval is a pumping interval. Stored A75 sources were used with the priority new API retrieval, then v1 XML, then flowtracing CSV, then entsoe-rest CSV; a lower-priority source is used only where those above have no value. The sources were retrieved at these times (UTC): v1 XML

Table 1: Input parameters per zone (MW). R, Q: Statnett queue, readings of 02.11 and 09.11 [2]. E: existing data-centre load [12, 13]; E_{2024} and E_{2025} are shown rounded to three decimals (the analysis uses the unrounded values of the v1 parameter file), and E_{2026} is given to three decimals. Reservoir and pumped storage: ENTSO-E A68 installed capacity for 2024 / 2025 / 2026 [11].

Zone	R (02.11)	Q (02.11)	R (09.11)	Q (09.11)	E_{2024}	E_{2025}	E_{2026}
NO1	1398	2220	1413	2220	40.605	96.967	106.676
NO2	1950	3092	1950	3668	17.978	26.684	62.196
NO3	96	1646	96	1745	93.707	153.827	109.066
NO4	477	2258	477	2248	70.993	81.096	77.291
NO5	270	894	270	894	0.891	0.057	0.039
Zone	Hydro Water Reservoir				Hydro Pumped Storage		
NO1	1419.81	1401.70	1401.60		n/e	n/e	n/e
NO2	9816.30	9870.80	9854.10		522.00	522.00	532.40
NO3	3415.56	3421.50	3448.00		60.20	60.20	61.20
NO4	4906.57	4914.30	4852.80		n/e	n/e	n/e
NO5	7101.92	7104.00	7123.90		482.50	482.50	482.50

2026-11-02T11:08:39Z to 2026-11-02T12:41:45Z; flowtracing CSV 2026-09-21T17:59:07Z to 2026-09-21T21:27:38Z; entsoe-rest CSV (file times) 2026-06-29T09:57:44Z to 2026-06-29T16:15:03Z; new API retrieval of A75 2026-11-09T17:26:37Z to 2026-11-09T18:04:34Z. In the overlaps between stored sources there are 0 deviating quarter-hours. Physical flows (ENTSO-E A11 [11]) are used only to check the sign of the net position; the A11 gap retrieval ran from 2026-11-09T18:04:55Z to 2026-11-09T19:50:37Z.

2.5 Existing data-centre load: Elhub

Existing data-centre load E per zone and year comes from the Elhub dataset `consumption_datacenter_per_mba_hour` (CC BY 4.0), downloaded 2026-11-02 [12]. For 2024 and 2025 we use the values in the v1 parameter file [13]; the reference year is 2025. For 2026 we apply v1's rule to January–October 2026 (7296 hours) in the same file (Table 1).

2.6 Data-centre queue: Statnett

The queue per zone (reserved capacity R and capacity queue Q) is read from Statnett's Power BI on 02.11.2026 (primary, as in v1) and on 09.11.2026 (sensitivity) [2] (Table 1). The national totals were 4 190 MW reserved and 10 109 MW in the queue on the first reading, and 4 205 MW and 10 774 MW on the second, which is not archived with a screenshot [2]; the zone values in Table 1 do not sum exactly to these totals.

3 Method

The methods were preregistered before any headroom or outcome was computed, in the signed git tag `prereg-v1` (tag object `d567d764`, commit `7f6a98b`). The preregistration was submitted to OSF Registries with an embargo until 31.12.2027 selected, `https://osf.io/vd9np` [14]. The git archive of the tag, the preregistration and KDF-0001 are timestamped with OpenTimestamps; after `ots upgrade` each carries Bitcoin block-header attestations at heights 975224 and 975226, but `ots verify` could not check them against the block headers, because no local Bitcoin node was available. The analysis follows the preregistration except for the deviations in Section 4.

Primary measure. The import headroom is $H(t) = \mathrm{NP}(t) - \mathrm{minNP}(t)$, and an MTU is an event when $L_f > H(t)$. The net position is net export: "Net position denotes net export" [10].

The Nordic capacity calculation methodology defines that "a positive net position is equivalent to export from, and a negative net position is equivalent to import into, the particular bidding zone" [9]. This definition is in the amendment proposal from 2026 [9], not in the approved 2020 text [8]; therefore the sign was checked against A11 before the analysis. The check is the Pearson correlation between NP and the A11 net export (sum of flows out minus flows in over the zone's border directions; an MTU is used only when all of them have values, see T3), per MTU at JAO resolution, all Norwegian zones pooled; a negative correlation would have stopped the analysis. MTUs with $H(t) < 0$ are kept and assessed like other MTUs.

Added load. The added load follows v1 [13]: $L_f = (E_{\text{ref}} - E_y) + b\,(f\,R_{\text{new}} + f\,Q)$ with $R_{\text{new}} = \max(0, R - E_{\text{ref}}/b)$, where R is reserved capacity and Q the capacity queue. The primary utilisation is $b = 0.70$ with the 02.11 queue. E_{ref} is E for the reference year 2025, and E_y is E for the calendar year of the MTU. The share f runs over v1's grid of 101 points [13]. At $f = 0$ the load is $L_0 = E_{\text{ref}} - E_y$, which can be negative, so the row $f = 0$ shows events with only the adjustment for existing load.

Secondary measure. The secondary measure adds regulable hydropower (installed minus actual) to $H(t)$ as an upper bound. Regulable hydropower is Hydro Water Reservoir plus Hydro Pumped Storage, installed from A68 minus actual from A75; run-of-river is excluded. Outages are not subtracted; the bound is therefore optimistic. An A75 interval without production for Hydro Pumped Storage counts as a pumping interval and is set to 0 before duration weighting only if A75 consumption for Hydro Pumped Storage is above 0 in the same interval and zone; otherwise it is a data gap and the MTU is missing for the secondary measure. A missing Hydro Water Reservoir interval also makes the MTU missing for the secondary measure. The rule applies only to the secondary measure; NO1 and NO4 have no pumped-storage term.

Hours and missing data. An hour is an event if at least one MTU in the hour is an event; if an MTU in the hour is missing, the hour is missing. MTUs without NP, minNP or a required ENTSO-E value are left out of the measure concerned and counted as missing; nothing is imputed. ENTSO-E XML with curve type A03 is decoded by holding the previous value until the next position.

Outcomes. Outcomes are reported per zone, per f, per calendar year and for the subperiods before and after the JAO MTU shift, separately and together: hours with an event, the share of event hours and the share of event MTUs, and the numbers of missing MTUs and hours. The share of event MTUs is reported per subperiod only, because the MTU length changes; part years are not scaled. Without a reliability standard for Norwegian zones in the source archive, outcomes are descriptive, with no threshold and no smallest f. Both measures are reported, and neither is chosen afterwards.

Sensitivities. Two sensitivities are run one at a time, with everything else as in the primary analysis: utilisation 1.0, applied both in R_{new} and in the added load, and the 09.11 queue.

Independent check. A separate model instance wrote its own code from the preregistration in the tag, the tagged ground truth and deviation A2, without opening our results, code, the `gt/` directory or `AVVIK_FRA_PREREG.md`, and recomputed every outcome table from the raw data; the tables agree cell by cell, with 0 differences. It found the same MTU shift date and the same sign correlation.

4 Deviations from the preregistration

This section translates the file `AVVIK_FRA_PREREG.md` in the repository without changing its content. Each deviation was recorded there with reason and time before it was carried out.

A1: Gate B without consent from JAO; results published with JAO as source. Decision by the author: JAO raw data and JAO time series are not published. Results are published (outcome tables and figures in the manuscript) with JAO given as source, together with code including retrieval scripts so that others can retrieve the data themselves. (Manuscript note, not part of `AVVIK_FRA_PREREG.md`: in this manuscript the source is given as the JAO Nordic Publication Tool, `https://publicationtool.jao.eu`; the JAO terms require users to "mention the Information Systems as the source of the Content, in accordance with good industry practices" [6].) Publication gate B no longer waits for written consent from JAO, and objections from JAO after publication would be handled with a correction; JAO is not contacted. The gate file that was part of the OSF registration required written consent from JAO; the requirement was removed in a separate commit, and the registered version is unchanged in the tag. The preregistration says that no JAO raw data or derived series are published before written consent from JAO. Raw data and time series are still not published, so that part of the rule stands. The deviation concerns the results: the JAO terms count derived material as content ("materials derived from, based on, generated from or incorporating the Content" [6]), and the outcome tables are derived from JAO data. Publishing them without consent therefore deviates from the preregistration to the extent the tables count as derived material. The open point on bots remains.

A2: JAO fbDomainShadowPrice replaces netPos as the source of NP. The preregistered source of NP was JAO Nordic `netPos`; `netPos` returned no data for 575 business days and data for 158, from 2026-05-28. By decision of the author, `fbDomainShadowPrice` (`cnecName` `NetPosition_<zone>`, field `flowFb`) replaces `netPos` for the whole period 2024-10-30 to 2026-10-31. Reason: the `netPos` page is built on the same dataset [5], and the two are identical in the overlap (75 380 pairs, 0 deviations). The deviation was recorded before the remaining days were retrieved and before the analysis was run; the 159 days from the reconnaissance, and the comparison with `netPos` that gave the reason for A2, came earlier the same day, and the 159 days were taken over unchanged (a clarification added to the file before publication). The retrieval followed the same rules, and the MTU shift was recomputed from the retrieved data and recorded before the analysis. A regression check required that event/non-event per MTU and per hour in the overlap be identical to the previous run; it found 0 MTUs and 0 hours with a different status. The rest of the preregistration applies unchanged.

A3: Gate B does not wait for OSF; the embargo stands. Decision by the author: gate B does not wait for OSF, and the embargo stands. Gate B began with the step of asking for the OSF embargo to be ended at least 48 hours before publication and confirming that the registration is public before the Zenodo publication; this step is dropped. The registration was submitted to OSF Registries with an embargo until 31.12.2027 selected, and the embargo stands; the manuscript does not call the embargo approved. The requirement in note M3 to update the embargo text in the method section when the embargo has been lifted still applies. The gate file is part of the OSF registration, so the change is recorded in `AVVIK_FRA_PREREG.md`.

Interpretations (not deviations). These are choices where the preregistration does not give a single reading; each was recorded before use.

- T1, outcome rows per year and subperiod: outcomes are reported in the cross of year (2024, 2025, 2026, all) and subperiod (before, after, together); combinations without MTUs are

left out, and the share of event MTUs is empty for the subperiod *together*.

- T2, days between the datasets' shift dates belong to the 60-minute subperiod; a dataset with 15-minute MTUs there is placed on the hour with duration weighting, requiring values in all four quarters.
- T3, A11 net export per MTU: the zone's border directions in the retrieval set, 30 in all; an MTU enters the correlation only when all of the zone's directions have values in all quarters of the MTU.
- T4, expected grid: the period [2024-10-30T00:00Z, 2026-11-01T00:00Z) with 60-minute MTUs up to the start of the shift day and 15-minute MTUs after; an MTU in the grid without NP or minNP is missing.
- T5, length of a CSV row: the smallest distance to the neighbouring rows' time stamps, at most 60 minutes; rows with a length other than 15 or 60 minutes are counted, and there were 0.
- T6, threshold for deviations in the overlap between stored sources: the same as in the revision check for A65 NO1.
- T7, calendar year: in UTC from the start of the MTU, as for A68; the same applies to E_y and the year rows.

Two notes from the data run, M1 and M2, concern `netPos` and the MTU-shift date before A2; with A2 both datasets shift on the same day, and the boundary is 2025-09-30T22:00Z. Note M3 adds to gate B the requirement to update the embargo text in the method section when the embargo has been lifted.

5 Results

The sign check is positive in every zone, and pooled over all zones the Pearson correlation is 0.954609 over 230 410 MTU pairs. The lowest zone value is 0.888775 in NO2. Tables 2 and 3 give the hours with an event per zone and year for selected f in the primary scenario; Figure 1 shows the whole f grid. Table 4 gives the shares for the subperiods before and after the MTU shift, and Table 5 the two sensitivities. Table 6 counts missing MTUs and hours, and MTUs with $H < 0$. Over the whole period, the primary measure has 17567 valid hours and 1 missing hour in every zone.

The secondary measure has fewer valid hours in NO2, NO3 and NO5 (Tables 3 and 6), because of pumped-storage data gaps: intervals in which no stored A75 source has a production value for Hydro Pumped Storage and none records consumption above 0. In 2025 this affects NO3 in 2025-10, with 13 such quarter-hours in both the v1 XML and the flowtracing CSV, giving 4 missing hours, and NO5 in 2025-11, with 5 such quarter-hours, giving 5 missing hours. These are the hours by which NO3 and NO5 fall short of the full year 2025 in Table 3. In 2026 the gaps lie in 2026-01 to 2026-05 in NO2 and 2026-03 to 2026-05 in NO3 and NO5, when only the entsoe-rest CSV, which lacks the consumption field, covers A75. MTUs with $H < 0$ occur only in NO5 (8 MTUs, all before the MTU shift), which is why NO5 has 8 event hours already at $f = 0$.

The numbers are counts of MTUs and hours in which the added load would exceed the published headroom. They are not estimates of unserved energy or of actual import needs.

6 Limitations

The limitations below are those of the preregistration, translated in full; the bullet on `netPos` is updated for deviation A2.

Table 2: Hours with an event, primary measure, primary scenario ($b = 0.70$, 02.11 queue), per zone and calendar year. Part years are not scaled.

Zone	Year	Hours with event, $f =$					Valid hours
		0.00	0.25	0.50	0.75	1.00	
NO1	2024	0	0	0	0	0	1512
	2025	0	0	24	176	594	8760
	2026	0	3	96	242	421	7295
	all	0	3	120	418	1015	17567
NO2	2024	0	0	0	0	5	1512
	2025	0	15	59	227	658	8760
	2026	0	0	26	204	761	7295
	all	0	15	85	431	1424	17567
NO3	2024	0	2	19	31	47	1512
	2025	0	61	196	312	646	8760
	2026	0	14	70	260	640	7295
	all	0	77	285	603	1333	17567
NO4	2024	0	23	231	404	601	1512
	2025	0	58	444	1242	3363	8760
	2026	0	0	27	333	1020	7295
	all	0	81	702	1979	4984	17567
NO5	2024	0	0	0	0	0	1512
	2025	8	13	18	25	31	8760
	2026	0	0	5	20	22	7295
	all	8	13	23	45	53	17567

Table 3: Hours with an event, secondary measure (optimistic upper bound), primary scenario, per zone and calendar year. Valid hours are fewer than in Table 2 where pumped-storage data gaps make MTUs missing.

Zone	Year	Hours with event, $f =$					Valid hours
		0.00	0.25	0.50	0.75	1.00	
NO1	2024	0	0	0	0	0	1512
	2025	0	0	0	0	38	8760
	2026	0	0	0	18	142	7295
	all	0	0	0	18	180	17567
NO2	2024	0	0	0	0	0	1512
	2025	0	0	0	0	0	8760
	2026	0	0	0	0	0	6445
	all	0	0	0	0	0	16717
NO3	2024	0	0	0	0	0	1512
	2025	0	0	0	0	0	8756
	2026	0	0	0	0	0	6651
	all	0	0	0	0	0	16919
NO4	2024	0	0	0	0	0	1512
	2025	0	0	0	0	0	8760
	2026	0	0	0	0	0	7295
	all	0	0	0	0	0	17567
NO5	2024	0	0	0	0	0	1512
	2025	0	0	0	0	0	8755
	2026	0	0	0	0	0	7014
	all	0	0	0	0	0	17281

Table 4: Subperiods before and after the JAO MTU shift, primary scenario: valid MTUs, share of event MTUs and share of event hours (%). The share of event MTUs is not pooled across the subperiods.

Measure	Zone	Before the MTU shift (60-min MTU)			After the MTU shift (15-min MTU)		
		valid MTU	% MTU	% hours	valid MTU	% MTU	% hours
primary measure, primary scenario, $f = 0.50$							
	NO1	8062	0.30	0.30	38020	0.88	1.01
	NO2	8062	0.66	0.66	38020	0.27	0.34
	NO3	8062	2.42	2.42	38020	0.83	0.95
	NO4	8062	4.80	4.80	38020	2.95	3.31
	NO5	8062	0.22	0.22	38020	0.04	0.05
primary measure, primary scenario, $f = 1.00$							
	NO1	8062	7.11	7.11	38020	4.48	4.65
	NO2	8062	7.54	7.54	38020	7.44	8.58
	NO3	8062	7.39	7.39	38020	7.31	7.75
	NO4	8062	36.01	36.01	38020	20.62	21.89
	NO5	8062	0.38	0.38	38020	0.20	0.23
secondary measure, primary scenario, $f = 0.50$							
	NO1	8062	0.00	0.00	38020	0.00	0.00
	NO2	8062	0.00	0.00	34855	0.00	0.00
	NO3	8062	0.00	0.00	35479	0.00	0.00
	NO4	8062	0.00	0.00	38020	0.00	0.00
	NO5	8062	0.00	0.00	36913	0.00	0.00
secondary measure, primary scenario, $f = 1.00$							
	NO1	8062	0.41	0.41	38020	1.48	1.55
	NO2	8062	0.00	0.00	34855	0.00	0.00
	NO3	8062	0.00	0.00	35479	0.00	0.00
	NO4	8062	0.00	0.00	38020	0.00	0.00
	NO5	8062	0.00	0.00	36913	0.00	0.00

Table 5: Sensitivities, run one at a time: hours with an event over the whole period. *utilisation 1.0* replaces b in both places; *queue read 09.11* replaces R and Q.

Measure	Scenario	Zone	Hours with event, $f =$				Valid hours
			0.25	0.50	0.75	1.00	
primary	primary	NO1	3	120	418	1015	17567
		NO2	15	85	431	1424	17567
		NO3	77	285	603	1333	17567
		NO4	81	702	1979	4984	17567
		NO5	13	23	45	53	17567
	utilisation 1.0	NO1	31	378	1396	4530	17567
		NO2	31	353	1898	4482	17567
		NO3	179	541	1576	2955	17567
		NO4	265	1752	6507	12128	17567
		NO5	16	42	54	64	17567
	queue read 09.11	NO1	3	123	420	1040	17567
		NO2	18	132	698	2205	17567
		NO3	93	305	708	1540	17567
		NO4	80	697	1953	4910	17567
		NO5	13	23	45	53	17567
secondary	primary	NO1	0	0	18	180	17567
		NO2	0	0	0	0	16717
		NO3	0	0	0	0	16919
		NO4	0	0	0	0	17567
		NO5	0	0	0	0	17281
	utilisation 1.0	NO1	0	13	281	1715	17567
		NO2	0	0	0	0	16717
		NO3	0	0	0	4	16919
		NO4	0	0	0	0	17567
		NO5	0	0	0	0	17281
	queue read 09.11	NO1	0	0	19	184	17567
		NO2	0	0	0	0	16717
		NO3	0	0	0	0	16919
		NO4	0	0	0	0	17567
		NO5	0	0	0	0	17281

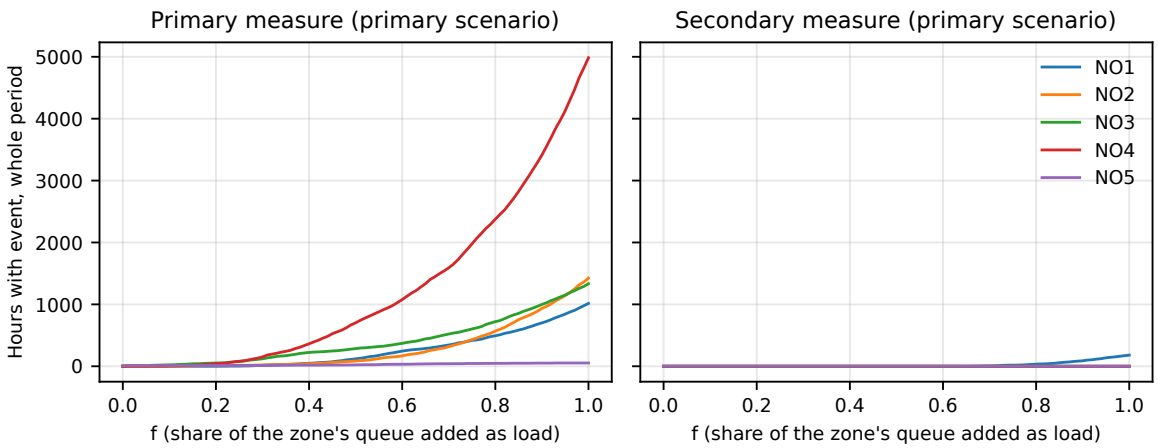


Figure 1: Hours with an event over the whole period against f, per zone, primary scenario. Left: primary measure. Right: secondary measure (optimistic upper bound).

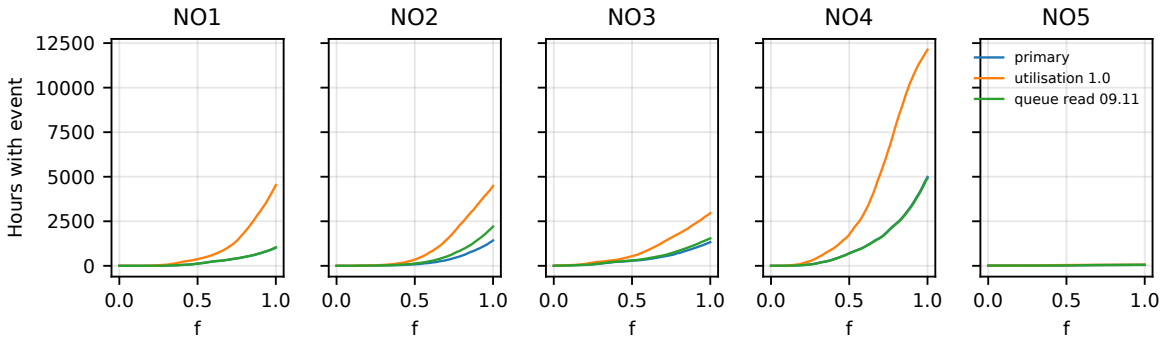


Figure 2: Primary measure: hours with an event against f per zone for the primary scenario and the two sensitivities.

- The extremes are not simultaneously feasible: "simultaneous feasibility of e.g. maximum flow on two borders cannot be guaranteed" [5]. Each minimum net position is a separate extreme of the domain. The result is an optimistic upper bound for available import, given that the domain does not move.
- The domain would move with large new load, and system properties, voltage and stability lie outside the method. Statnett gave system properties as the reason for the halt [1], and § 14 of the Norwegian regulation on the system responsibility (fos) concerns approval of plant functionality [1].
- Queues in other zones are left out.
- Queue figures vary with source and date: the press release gives 4 200 MW reserved and 9 100 MW in the queue [1]; the Power BI reading of 02.11 gives 4 190 and 10 109, which is the queue of the primary analysis [2]; the reading of 09.11 gives 4 205 and 10 774 and is not archived with a screenshot [2]; the long-term market analysis gives about 4200 and about 8800 [4].
- Reserved and connected capacity: when a project is connected, its status changes from reserved to connected, but some connected projects with actual power exchange are still listed as reserved, because the cases are updated at least once a year [2]. Statnett's reserved figures can therefore include projects already connected, so the same capacity can be

Table 6: Missing MTUs and hours per zone and measure over the whole period, and MTUs with $H < 0$ (primary measure).

Zone	MTU in grid	Missing MTU		Hours in grid	Missing hours		MTU with $H < 0$
		primary	secondary		primary	secondary	(primary)
NO1	46086	4	4	17568	1	1	0
NO2	46086	4	3169	17568	1	851	0
NO3	46086	4	2545	17568	1	649	0
NO4	46086	4	4	17568	1	1	0
NO5	46086	4	1111	17568	1	287	8

counted both in the net position and in the queue; events may then be overestimated, by an unknown amount.

- The last year of the period uses E_y for January–October, while 2024 and 2025 use full-year values from the v1 parameter file.
- The preregistration noted that `netPos` gave no data for the first delivery day; in the data run `netPos` had no data for 575 business days, and with deviation A2 the realised net position comes from `fbDomainShadowPrice`, which covers the whole period.
- The secondary measure is an optimistic upper bound: outages are not subtracted.

7 Data and code availability

No JAO raw data and no JAO time series are published. The JAO terms define the Information Systems as "any electronic system, application, database, website, API, or any other digital platform, including but not limited to the JAO Website, the IIP Website and all its subpages and related links, that are owned, operated, or controlled by JAO" [6]. Section A of the terms carries the heading "GENERAL PROVISIONS APPLICABLE TO ALL USE OF THE INFORMATION SYSTEMS" [6]. Under the clause on access and use, "When accessing the Information Systems and using the Content, a User shall", among other things, "mention the Information Systems as the source of the Content, in accordance with good industry practices" [6]. The clause on restrictions on use has two provisions on passing content on [6]. The first reads: "The Content cannot be sold, commercialized as a product or service, marketed or otherwise made available to any third party, in whole or in part, in any manner, freely or against payment, unless expressly authorized in advance by JAO in writing" [6]. The second reads: "The Content cannot be disseminated, displayed, copied, extracted, distributed or reproduced in whole or in part, in any manner, on any medium or support whatsoever, freely or against payment, unless expressly authorized in advance by JAO in writing, or unless it is consistent with a Legitimate Use or otherwise permitted by these Terms of Use" [6]. "Legitimate Use" is defined as any of a list of uses, of which the first is "Use for educational, academic or research purposes" [6]. For the purposes of the restrictions, content includes "materials derived from, based on, generated from or incorporating the Content" [6]. The outcome tables and figures in this manuscript are published with the JAO Nordic Publication Tool (`https://publicationtool.jao.eu`) as source (deviation A1). Code, including the retrieval scripts, is published so that others can retrieve the data themselves; after the data run the scripts were given a fixed pause between calls and a wait after HTTP 429, whereas the data run itself used no pause. ENTSO-E and Elhub data are used under their own terms; the Elhub dataset is CC BY 4.0 [12]. All sources are listed in the supplement `sources.csv` [15].

8 Use of AI tools

The analysis was planned and carried out with Claude Code as a tool under the author's direction. Tools and models, read from the session logs: Claude Code 2.1.317, main session on `claude-opus-5-5`; `claude-sonnet-5-5` gave 17 of the main session's answers, all in the interval 2026-11-09T16:27:15.690Z to 2026-11-09T16:27:46.448Z. Subagents in Claude Code, all on `claude-opus-5-5`: 4 for source reading, 23 for blind verification of ground-truth rows, 4 for independent review of the preregistration, and 2 for the independent recomputation of the outcomes; 33 in all, counted when this manuscript was written (later subagents are recorded in the repository). Chat-Claude (claude.ai) was used for the author's own review; according to the author, it ran Claude Opus 5.5 (`claude-opus-5-5`) throughout, without a change of model; this is the author's statement, archived as source K-115 in `sources.csv` [15], not a reading of the session logs. "The author made the methodological choices, had every figure checked against its source, and is responsible for the content."

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